

Laboratory Assignment 6: Introduction to Field Programmable Gate Arrays

ECE 0201: Digital Circuits and Systems

(45 Points)

Name

张鹤扬 Stefan

Submission Checklist:

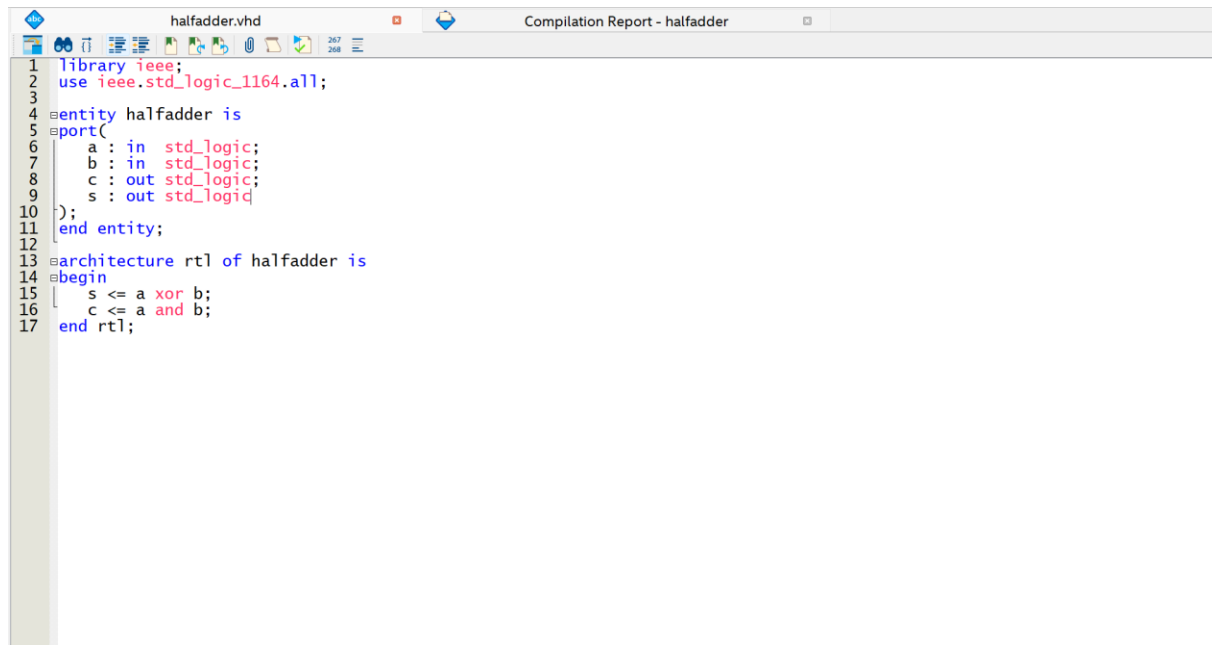
- ☐ Write within boxes, do not move boxes
- ☐ Write your full name in the box above

Save this file as a PDF before uploading, keep the number of pages (8) unchanged

Note “TO BE CONTINUED” in the answer box if you used the extra pages (7-8)

Part I: Tutorials (5 points)

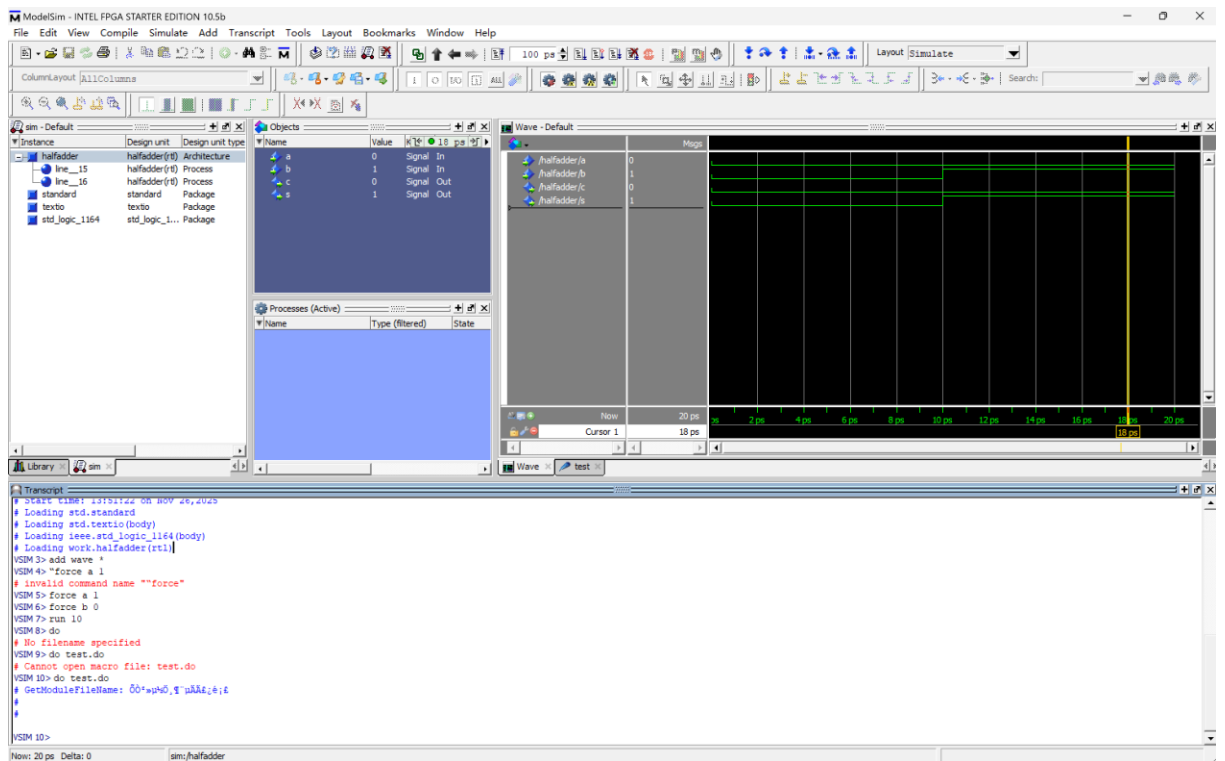
[(A) Insert a screenshot of your VHDL code for Part I.] (1 points)



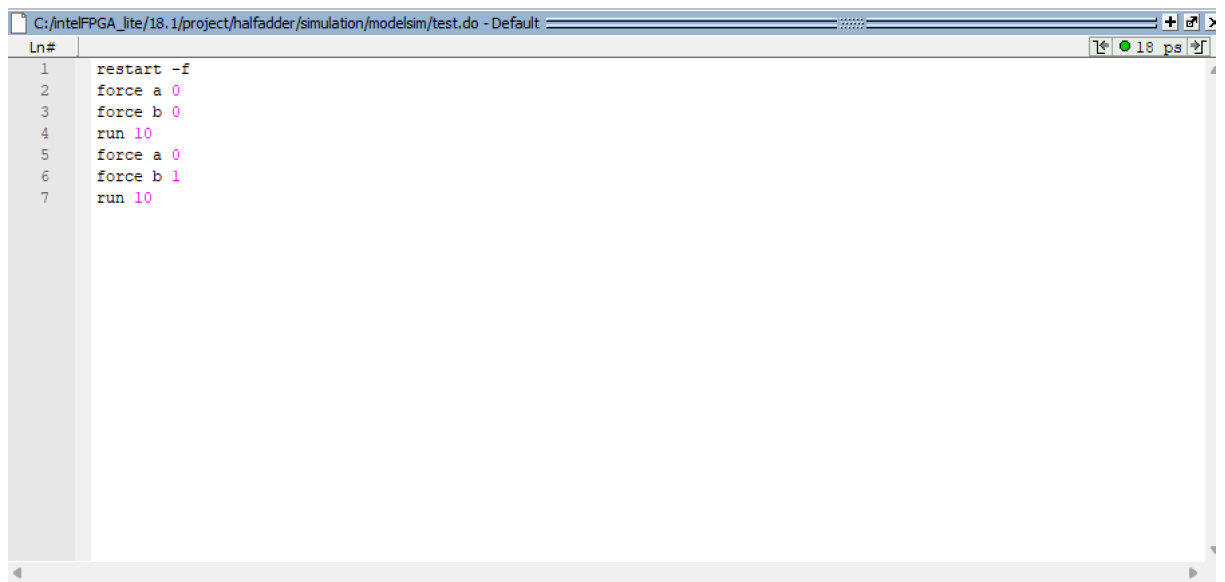
The screenshot shows a VHDL code editor window titled 'halfadder.vhd'. The code defines a half adder entity with two inputs (a, b) and two outputs (c, s). The architecture 'rtl' implements the logic using XOR and AND gates.

```
1 library ieee;
2 use ieee.std_logic_1164.all;
3
4 entity halfadder is
5 port(
6     a : in std_logic;
7     b : in std_logic;
8     c : out std_logic;
9     s : out std_logic
10 );
11 end entity;
12
13 architecture rtl of halfadder is
14 begin
15     s <= a xor b;
16     c <= a and b;
17 end rtl;
```

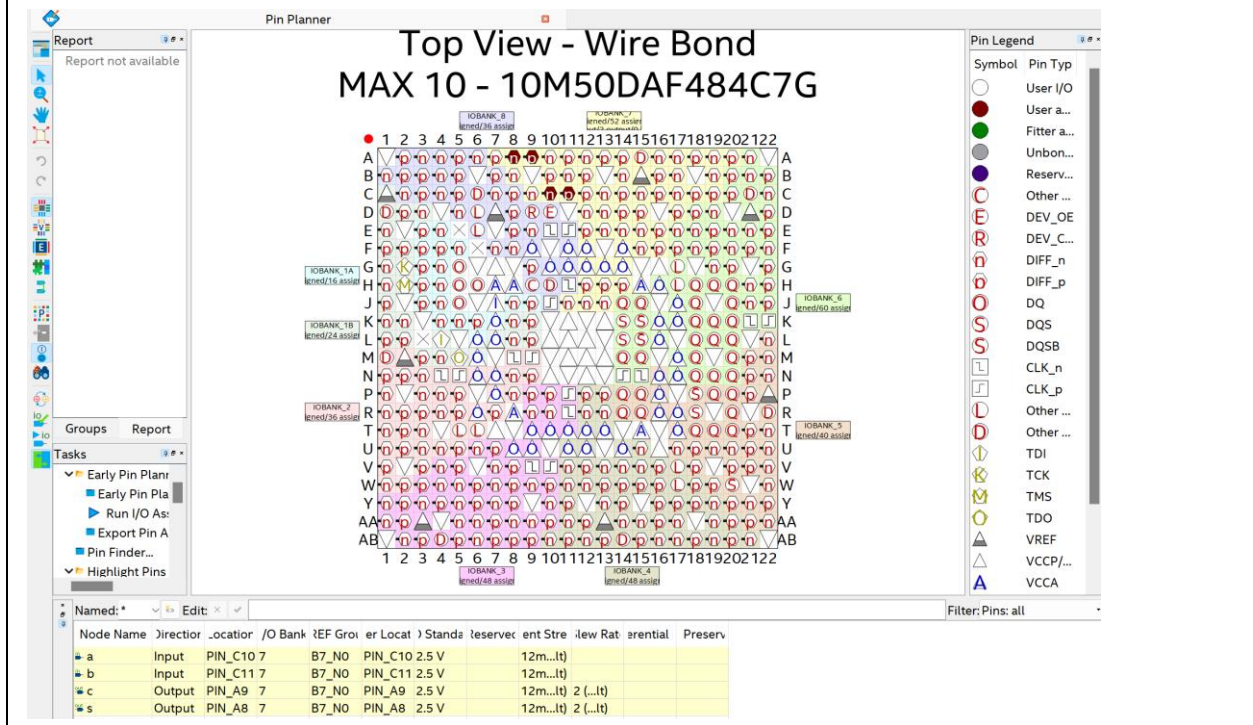
[(B) Insert an image of your ModelSim Simulation.] (2 points)



[(C) Insert the text of your do file.] (1 points)

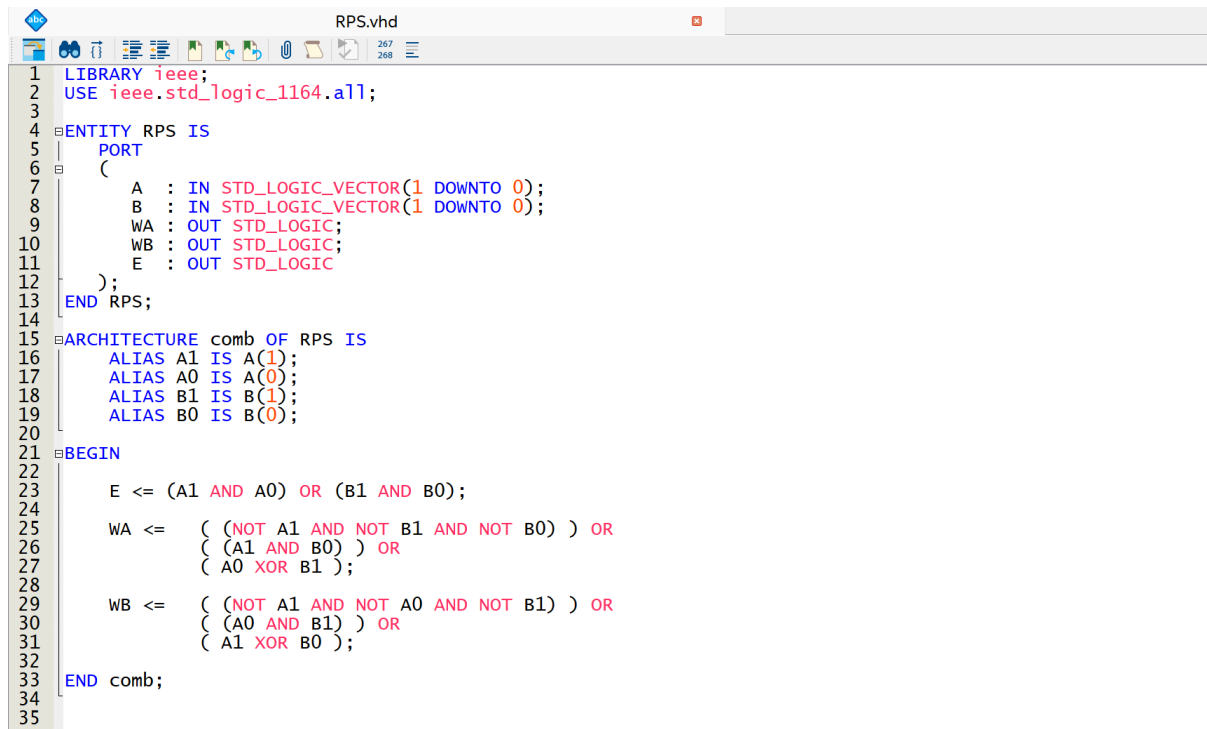


[(D) Insert a screenshot of the Pin Planner.] (1 point)



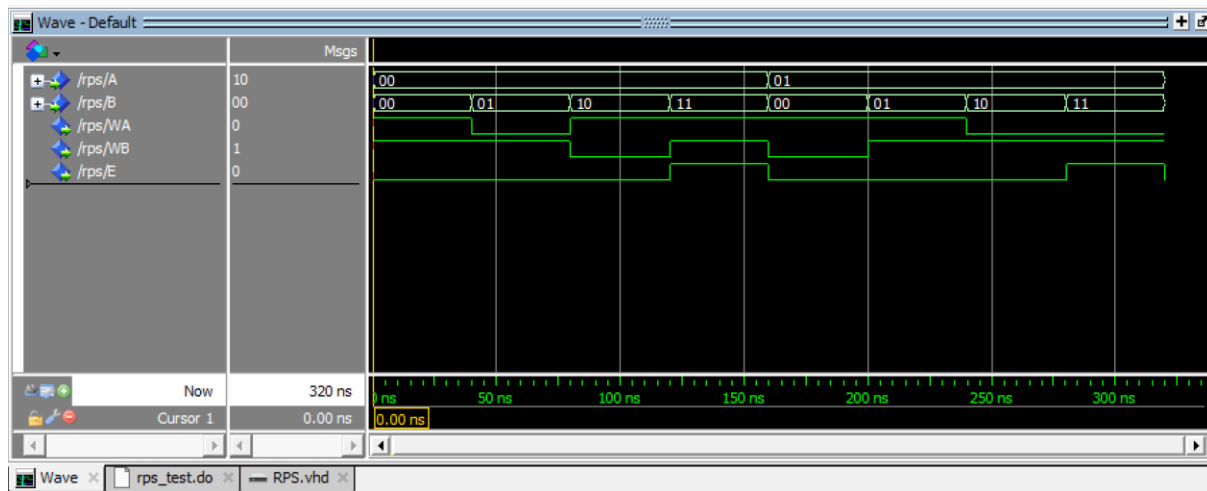
Part II: Rock-Paper-Scissors! (40 points)

[(A) Insert a screenshot of your VHDL code for Part II.] (4 points)



```
1  LIBRARY ieee;
2  USE ieee.std_logic_1164.all;
3
4  ENTITY RPS IS
5      PORT
6      (
7          A : IN STD_LOGIC_VECTOR(1 DOWNTO 0);
8          B : IN STD_LOGIC_VECTOR(1 DOWNTO 0);
9          WA : OUT STD_LOGIC;
10         WB : OUT STD_LOGIC;
11         E : OUT STD_LOGIC
12     );
13 END RPS;
14
15 ARCHITECTURE comb OF RPS IS
16     ALIAS A1 IS A(1);
17     ALIAS A0 IS A(0);
18     ALIAS B1 IS B(1);
19     ALIAS B0 IS B(0);
20
21 BEGIN
22
23     E <= (A1 AND A0) OR (B1 AND B0);
24
25     WA <= ( (NOT A1 AND NOT B1 AND NOT B0) ) OR
26           ( (A1 AND B0) ) OR
27           ( A0 XOR B1 );
28
29     WB <= ( (NOT A1 AND NOT A0 AND NOT B1) ) OR
30           ( (A0 AND B1) ) OR
31           ( A1 XOR B0 );
32
33 END comb;
```

[(B) Insert an image of your ModelSim Simulation.] (4 points)



[(C) Insert the text of your do file.] (2 points)

```
C:/intelFPGA_lite/18.1/project/RPS/simulation/modelsim/rps_test.do - Default
Ln#
1 onerror resume
2 vdel work
3 vlib work
4 vcom C:/intelFPGA_lite/18.1/project/RPS/RPS.vhd
5 vsim work.RPS
6 add wave *
7 config wave -timelineunits ns
8 view wave
9 set time 0
10 set step 20
11 proc dec_to_bin_2bit {dec} {
12     set bit1 [expr {($dec / 2) % 2}]
13     set bit0 [expr {$dec % 2}]
14     return "$bit1$bit0"
15 }
16 for {set A_val 0} {$A_val < 4} {incr A_val} {
17     set A_str [dec_to_bin_2bit $A_val]
18     for {set B_val 0} {$B_val < 4} {incr B_val} {
19         set B_str [dec_to_bin_2bit $B_val]
20         force /rps/A $A_str $time ns
21         force /rps/B $B_str $time ns
22         echo "Testing: A=$A_str vs B=$B_str"
23         set time [expr $time + $step]
24         run $step ns
25     }
26 }
27 wave zoom full
28 echo "--- Full 16-case test complete ---"
```

[(D) Demonstrate your FPGA implementation to the TA for Lab 6 check-off.] (30 points)

[Leave the page unchanged if you are not going to use. Otherwise, specify the problem number you are working on here. Don't forget to notify "TO BE CONTINUED" in the original problem box]

[Leave the page unchanged if you are not going to use. Otherwise, specify the problem number you are working on here. Don't forget to notify "TO BE CONTINUED" in the original problem box]